Name
Date
Course

Solving Systems of Equations

Method I: Substitution

① $\begin{cases} y = 2x \\ 5x + y = -14 \end{cases}$

$5x + 2x = -14$

$7x = -14$

$\boxed{x = -2}$

$y = 2(-2)$

$\boxed{y = -4}$

② $\begin{cases} x + y = 4 \\ x = 2 + y \end{cases}$

$2 + y + y = 4$

$2 + 2y = 4$

$\boxed{y = 1}$

$x = 2 + 1$

$\boxed{x = 3}$

③ $\begin{cases} -6x = y \\ -x + y = 5 \end{cases}$

$-x - 6x = 5$

$-7x = 5$

$\boxed{x = -\frac{5}{7}}$

$-6(-\frac{5}{7}) = y$

$-\frac{6}{1}(-\frac{5}{7}) = y$

$\boxed{\frac{30}{7} = y}$

④ $\begin{cases} x + y = 5 \\ y = x + 1 \end{cases}$

$x + x + 1 = 5$

$2x + 1 = 5$

$\boxed{x = 2}$

$y = 2 + 1$

$\boxed{y = 3}$

⑤ $\begin{cases} 4y = x \\ 2y + x = 12 \end{cases}$

$2y + 4y = 12$

$6y = 12$

$\boxed{y = 2}$

$4(2) = x$

$\boxed{8 = x}$

⑥ $\begin{cases} y - 3x = 1 \\ -1 + x = y \end{cases}$

$-1 + x - 3x = 1$

$-1 - 2x = 1$

$\boxed{x = -1}$

$-1 - 1 = y$

$\boxed{-2 = y}$

⑦ $\begin{cases} 2y + 1 = x \\ x - y = 4 \end{cases}$

$2y + 1 - y = 4$

$y + 1 = 4$

$\boxed{y = 3}$

$2(3) + 1 = x$

$6 + 1 = x$

$\boxed{7 = x}$

⑧ $\begin{cases} 2x - 1 = y \\ y + 7x = 4 \end{cases}$

$2x - 1 + 7x = 4$

$9x - 1 = 4$

$\boxed{x = \frac{5}{9}}$

$2(\frac{5}{9}) - 1 = y$

$\frac{2}{1}(\frac{5}{9}) - 1 = y$

$\frac{10}{9} - 1 = y$

$\frac{10}{9} - \frac{9}{9} = y$

$\boxed{\frac{1}{9} = y}$

⑨ $\begin{cases} y + 2x = 7 \\ x = 3y \end{cases}$

$y + 2(3y) = 7$

$y + 6y = 7$

$7y = 7$

$\boxed{y = 1}$

$x = 3(1)$

$\boxed{x = 3}$

⑩ $\begin{cases} -3y + 8x = 1 \\ -4x = y \end{cases}$

$-3(-4x) + 8x = 1$

$12x + 8x = 1$

$20x = 1$

$\boxed{x = \frac{1}{20}}$

$-4(\frac{1}{20}) = y$

$-\frac{4}{1}(\frac{1}{20}) = y$

$-\frac{4}{20} = y$

$\boxed{-\frac{1}{5} = y}$

⑪ $\begin{cases} 5y = x \\ 7y - 2x = 6 \end{cases}$

$7y - 2(5y) = 6$

$7y - 10y = 6$

$-3y = 6$

$\boxed{y = -2}$

$5(-2) = x$

$\boxed{-10 = x}$

⑫ $\begin{cases} -8x - 7y = 4 \\ y = -9x \end{cases}$

$-8x - 7(-9x) = 4$

$-8x + 63x = 4$

$55x = 4$

$\boxed{x = \frac{4}{55}}$

$y = -9(\frac{4}{55})$

$y = -\frac{9}{1}(\frac{4}{55})$

$\boxed{y = -\frac{36}{55}}$

13. $\begin{cases} -3x+2y=4 \\ y=2x+1 \end{cases}$

$$-3x+2(2x+1)=4$$
$$-3x+4x+2=4$$
$$x+2=4$$
$$\boxed{x=2}$$
$$y=2(2)+1$$
$$y=4+1$$
$$\boxed{y=5}$$

14. $\begin{cases} 3y-4=x \\ -2x+7y=6 \end{cases}$

$$-2(3y-4)+7y=6$$
$$-6y+8+7y=6$$
$$8+y=6$$
$$\boxed{y=-2}$$
$$3(-2)-4=x$$
$$-6-4=x$$
$$\boxed{-10=x}$$

15. $\begin{cases} -4x-1=y \\ x-2y=5 \end{cases}$

$$x-2(-4x-1)=5$$
$$x+8x+2=5$$
$$9x+2=5$$
$$x=\frac{3}{9}$$
$$\boxed{x=\frac{1}{3}}$$
$$-4\left(\frac{1}{3}\right)-1=y$$
$$-\frac{4}{3}-\frac{1}{1}=y$$
$$-\frac{4}{3}-\frac{3}{3}=y$$
$$\boxed{-\frac{7}{3}=y}$$

16. $\begin{cases} x=3y-2 \\ -3x+7y=8 \end{cases}$

$$-3(3y-2)+7y=8$$
$$-9y+6+7y=8$$
$$-2y+6=8$$
$$\boxed{y=-1}$$
$$x=3(-1)-2$$
$$x=-3-2$$
$$\boxed{x=-5}$$

17. $\begin{cases} -3x+2y=11 \\ y=5x+2 \end{cases}$

$$-3x+2(5x+2)=11$$
$$-3x+10x+4=11$$
$$7x+4=11$$
$$\boxed{x=1}$$
$$y=5(1)+2$$
$$y=5+2$$
$$\boxed{y=7}$$

18. $\begin{cases} 3y-x=-1 \\ 2y-4=x \end{cases}$

$$3y-(2y-4)=-1$$
$$3y-1(2y-4)=-1$$
$$3y-2y+4=-1$$
$$y+4=-1$$
$$\boxed{y=-5}$$
$$2(-5)-4=x$$
$$-14=x$$

19. $\begin{cases} 1-4x=y \\ x-y=4 \end{cases}$

$$x-(1-4x)=4$$
$$x-1(1-4x)=4$$
$$x-1+4x=4$$
$$-1+5x=4$$
$$\boxed{x=1}$$
$$1-4(1)=y$$
$$1-4=y$$
$$\boxed{-3=y}$$

20. $\begin{cases} 8=-y-x \\ x=-7+2y \end{cases}$

$$8=-y-(-7+2y)$$
$$8=-y-1(-7+2y)$$
$$8=-y+7-2y$$
$$8=-3y+7$$
$$\boxed{-\frac{1}{3}=y}$$
$$x=-7+2\left(-\frac{1}{3}\right)$$
$$x=-\frac{7}{1}-\frac{2}{3}$$
$$x=-\frac{21}{3}-\frac{2}{3}=\boxed{-\frac{23}{3}}$$

21. $\begin{cases} -2+9x=y \\ x-y=10 \end{cases}$

$$x-(-2+9x)=10$$
$$x-1(-2+9x)=10$$
$$x+2-9x=10$$
$$-8x+2=10$$
$$\boxed{x=-1}$$
$$-2+9(-1)=y$$
$$-2-9=y$$
$$\boxed{-11=y}$$

Method II: Elimination

22. $\begin{cases} -2x+y=-3 \\ 2x+3y=7 \end{cases}$

$$4y=4$$
$$\boxed{y=1}$$
$$-2x+1=-3$$
$$\boxed{x=2}$$

23. $\begin{cases} 2x+3y=5 \\ x-3y=-2 \end{cases}$

$$3x=3$$
$$\boxed{x=1}$$
$$1-3y=-2$$
$$\boxed{y=1}$$

24. $\begin{cases} x-2y=1 \\ -x+7y=-6 \end{cases}$

$$5y=-5$$
$$\boxed{y=-1}$$
$$x-2(-1)=1$$
$$x+2=1$$
$$\boxed{x=-1}$$

(25) $\begin{cases} -5x+y=-4 \\ 5x+2y=7 \end{cases}$

$3y=3$

$\boxed{y=1}$

$5x+2(1)=7$

$5x+2=7$

$\boxed{x=1}$

(26) $\begin{cases} 2x+7y=5 \\ 3x-7y=-10 \end{cases}$

$5x=-5$

$\boxed{x=-1}$

$2(-1)+7y=5$

$-2+7y=5$

$\boxed{y=1}$

(27) $\begin{cases} 9x-y=7 \\ 2x+y=4 \end{cases}$

$11x=11$

$\boxed{x=1}$

$2(1)+y=4$

$2+y=4$

$\boxed{y=2}$

(28) $\begin{cases} 10x+9y=20 \\ -10x+8y=14 \end{cases}$

$17y=34$

$\boxed{y=2}$

$10x+9(2)=20$

$10x+18=20$

$x=\frac{2}{10}$

$\boxed{x=\frac{1}{5}}$

(29) $\begin{cases} 4x+8y=0 \\ -3x-8y=-4 \end{cases}$

$\boxed{x=-4}$

$4(-4)+8y=0$

$-16+8y=0$

$\boxed{y=2}$

(30) $\begin{cases} 4x+y=5 \\ 3x-2y=1 \end{cases}$

$\begin{cases} 2(4x+y)=2(5) \\ 3x-2y=1 \end{cases}$

$\begin{cases} 8x+2y=10 \\ 3x-2y=1 \end{cases}$

$11x=11$

$\boxed{x=1}$

$3(1)-2y=1$

$\boxed{y=1}$

(31) $\begin{cases} -6x+11y=-1 \\ 2x-3y=1 \end{cases}$

$\begin{cases} -6x+11y=-1 \\ 3(2x-3y)=3(1) \end{cases}$

$\begin{cases} -6x+11y=-1 \\ 6x-9y=3 \end{cases}$

$2y=2$

$\boxed{y=1}$

$2x-3(1)=1$

$2x-3=1$

$\boxed{x=2}$

(32) $\begin{cases} 5x+y=6 \\ -10x+3y=-7 \end{cases}$

$\begin{cases} 2(5x+y)=2(6) \\ -10x+3y=-7 \end{cases}$

$\begin{cases} 10x+2y=12 \\ -10x+3y=-7 \end{cases}$

$5y=5$

$\boxed{y=1}$

$5x+1=6$

$\boxed{x=1}$

(33) $\begin{cases} 3x+y=3 \\ 2x-3y=13 \end{cases}$

$\begin{cases} 3(3x+y)=3(3) \\ 2x-3y=13 \end{cases}$

$\begin{cases} 9x+3y=9 \\ 2x-3y=13 \end{cases}$

$11x=22$

$\boxed{x=2}$

$2(2)-3y=13$

$\boxed{y=-3}$

(34) $\begin{cases} 2x+5y=1 \\ -4x+3y=11 \end{cases}$

$\begin{cases} 2(2x+5y)=2(1) \\ -4x+3y=11 \end{cases}$

$\begin{cases} 4x+10y=2 \\ -4x+3y=11 \end{cases}$

$13y=13$

$\boxed{y=1}$

$2x+5(1)=1$

$2x+5=1$

$\boxed{x=-2}$

(35) $\begin{cases} -15x+7y=8 \\ 5x+4y=-9 \end{cases}$

$\begin{cases} -15x+7y=8 \\ 3(5x+4y)=3(-9) \end{cases}$

$\begin{cases} -15x+7y=8 \\ 15x+12y=-27 \end{cases}$

$19y=-19$

$\boxed{y=-1}$

$5x+4(-1)=-9$

$5x-4=-9$

$\boxed{x=-1}$

(36) $\begin{cases} 2x+5y=-1 \\ x+y=1 \end{cases}$

$\begin{cases} 2x+5y=-1 \\ -2(x+y)=-2(1) \end{cases}$

$\begin{cases} 2x+5y=-1 \\ -2x-2y=-2 \end{cases}$

$3y=-3$

$\boxed{y=-1}$

$x-1=1$

$\boxed{x=2}$

37. $\begin{cases} x+y=3 \\ 2x+3y=8 \end{cases}$

$\begin{cases} -3(x+y)=-3(3) \\ 2x+3y=8 \end{cases}$

$\begin{cases} -3x-3y=-9 \\ 2x+3y=8 \end{cases}$

$-x=-1$

$-1x=-1$

$\boxed{x=1}$

$1+y=3$

$\boxed{y=2}$

38. $\begin{cases} -9x+8y=-1 \\ -3x+5y=-5 \end{cases}$

$\begin{cases} -9x+8y=-1 \\ -3(-3x+5y)=-3(-5) \end{cases}$

$\begin{cases} -9x+8y=-1 \\ 9x-15y=15 \end{cases}$

$-7y=14$

$\boxed{y=-2}$

$-3x+5(-2)=-5$

$-3x-10=-5$

$\boxed{x=-\frac{5}{3}}$

39. $\begin{cases} 4x-7y=5 \\ 8x-5y=19 \end{cases}$

$\begin{cases} -2(4x-7y)=-2(5) \\ 8x-5y=19 \end{cases}$

$\begin{cases} -8x+14y=-10 \\ 8x-5y=19 \end{cases}$

$9y=9$

$\boxed{y=1}$

$8x-5(1)=19$

$8x-5=19$

$\boxed{x=3}$

40. $\begin{cases} 6x+7y=-5 \\ 5x+y=-9 \end{cases}$

$\begin{cases} 6x+7y=-5 \\ -7(5x+y)=-7(-9) \end{cases}$

$\begin{cases} 6x+7y=-5 \\ -35x-7y=63 \end{cases}$

$-29x=58$

$\boxed{x=-2}$

$5(-2)+y=-9$

$-10+y=-9$

$\boxed{y=1}$

41. $\begin{cases} -6x+5y=-1 \\ -2x-3y=-5 \end{cases}$

$\begin{cases} -6x+5y=-1 \\ -3(-2x-3y)=-3(-5) \end{cases}$

$\begin{cases} -6x+5y=-1 \\ 6x+9y=15 \end{cases}$

$14y=14$

$\boxed{y=1}$

$-6x+5(1)=-1$

$\boxed{x=1}$

42. $\begin{cases} 2x+3y=1 \\ -5x+4y=9 \end{cases}$

$\begin{cases} 5(2x+3y)=5(1) \\ 2(-5x+4y)=2(9) \end{cases}$

$\begin{cases} 10x+15y=5 \\ -10x+8y=18 \end{cases}$

$23y=23$

$\boxed{y=1}$

$2x+3(1)=1$

$\boxed{x=-1}$

43. $\begin{cases} 9x-5y=-1 \\ 8x+3y=14 \end{cases}$

$\begin{cases} 3(9x-5y)=3(-1) \\ 5(8x+3y)=5(14) \end{cases}$

$\begin{cases} 27x-15y=-3 \\ 40x+15y=70 \end{cases}$

$67x=67$

$\boxed{x=1}$

$8(1)+3y=14$

$8+3y=14$

$\boxed{y=2}$

44. $\begin{cases} 15x+8y=7 \\ 10x+5y=5 \end{cases}$

$\begin{cases} -5(15x+8y)=-5(7) \\ 8(10x+5y)=8(5) \end{cases}$

$\begin{cases} -75x-40y=-35 \\ 80x+40y=40 \end{cases}$

$5x=5$

$\boxed{x=1}$

$10(1)+5y=5$

$10+5y=5$

$\boxed{y=-1}$

45. $\begin{cases} 3x+7y=4 \\ -5x+2y=7 \end{cases}$

$\begin{cases} 5(3x+7y)=5(4) \\ 3(-5x+2y)=3(7) \end{cases}$

$\begin{cases} 15x+35y=20 \\ -15x+6y=21 \end{cases}$

$41y=41$

$\boxed{y=1}$

$3x+7(1)=4$

$3x+7=4$

$\boxed{x=-1}$

46. $\begin{cases} 6x-11y=5 \\ 8x+3y=-11 \end{cases}$

$\begin{cases} 3(6x-11y)=3(5) \\ 11(8x+3y)=11(-11) \end{cases}$

$\begin{cases} 18x-33y=15 \\ 88x+33y=-121 \end{cases}$

$106x=-106$

$\boxed{x=-1}$

$6(-1)-11y=5$

$-6-11y=5$

$\boxed{y=-1}$

47. $\begin{cases} 7x+3y=1 \\ 4x+7y=-10 \end{cases}$

$\begin{cases} -4(7x+3y)=-4(1) \\ 7(4x+7y)=7(-10) \end{cases}$

$\begin{cases} -28x-12y=-4 \\ 28x+49y=-70 \end{cases}$

$37y=-74$

$\boxed{y=-2}$

$7x+3(-2)=1$

$7x-6=1$

$\boxed{x=1}$

Challenge Problems

48) $\begin{cases} 2x+3y=1 \\ 5x+4y=1 \end{cases}$

49) $\begin{cases} 6x+5y=-1 \\ 4x+7y=-1 \end{cases}$

50) $\begin{cases} 5x+6y=4 \\ 8x+9y=6 \end{cases}$

$\begin{cases} -5(2x+3y)=-5(1) \\ 2(5x+4y)=2(1) \end{cases}$

$\begin{cases} -10x-15y=-5 \\ 10x+8y=2 \end{cases}$

$-7y=-3$

$\boxed{y=\frac{3}{7}}$

$2x+3\left(\frac{3}{7}\right)=1$

$2x+\frac{9}{7}=1$

$2x=1-\frac{9}{7}$

$2x=\frac{1}{1}-\frac{9}{7}$

$2x=\frac{7}{7}-\frac{9}{7}$

$2x=-\frac{2}{7}$

$\frac{2x}{2}=\frac{-\frac{2}{7}}{2}$

$x=-\frac{2}{7}\div\frac{2}{1}$

$x=-\frac{2}{7}\cdot\frac{1}{2}$

$x=-\frac{2}{7}\cdot\frac{1}{2}$

$x=-\frac{2}{14}$

$\boxed{x=-\frac{1}{7}}$

51) $\begin{cases} 4x+2y=-1 \\ 3x+5y=1 \end{cases}$

52) $\begin{cases} 12x+5y=7 \\ 8x+9y=-1 \end{cases}$

53) $\begin{cases} 2x+6y=-1 \\ 3x+5y=1 \end{cases}$